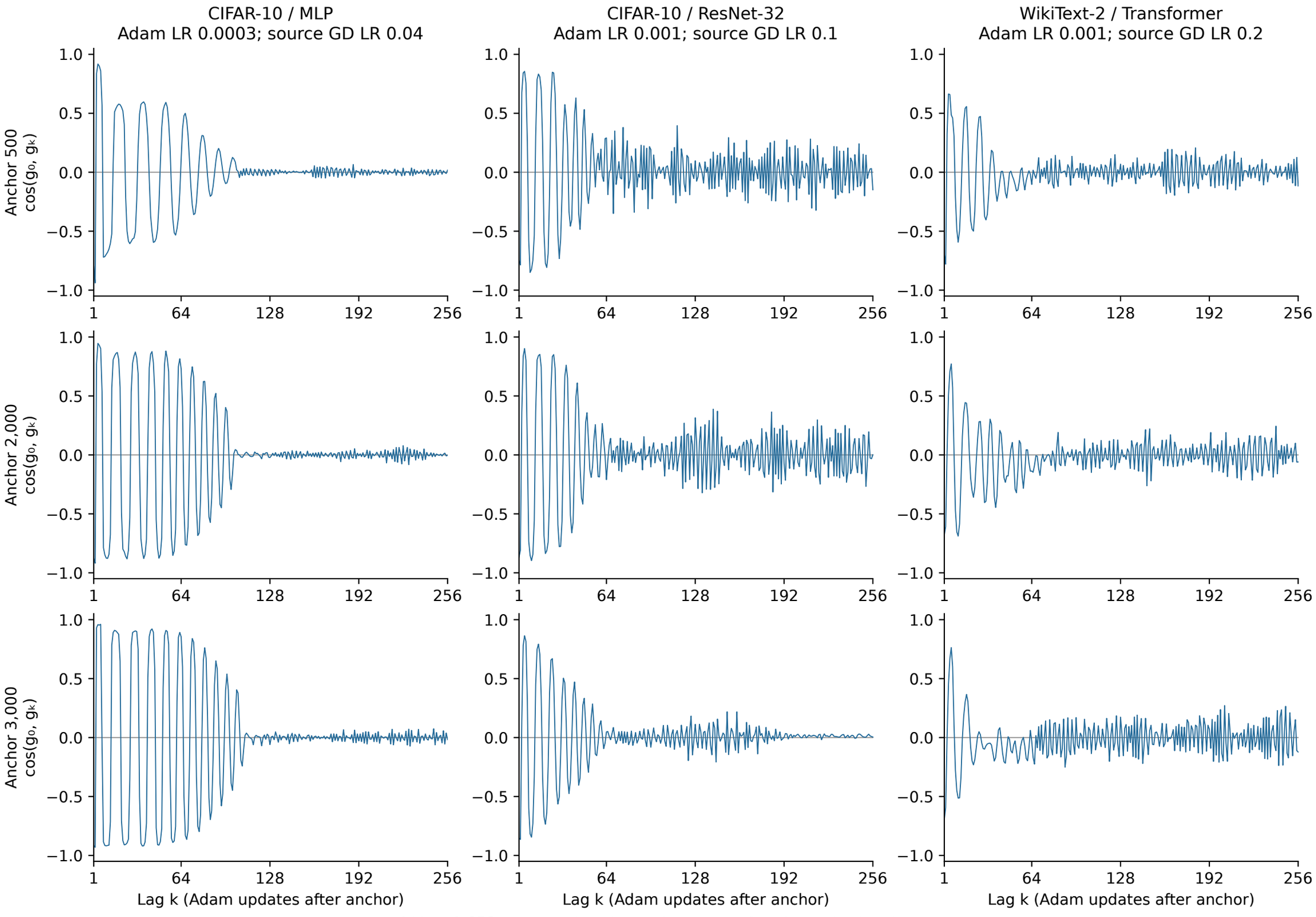

Anchored raw-gradient cosine along Adam from largest-rate GD states
CIFAR-10 / MLP
Adam LR 0.0003; source GD LR 0.04
CIFAR-10 / ResNet-32
Adam LR 0.001; source GD LR 0.1
WikiText-2 / Transformer
Adam LR 0.001; source GD LR 0.2
Anchor 500
Anchor 2,000
Anchor 3,000
cos(g₀, gₖ)
1.0
0.5
0.0
−0.5
−1.0
1
64
128
192
256
Lag k (Adam updates after anchor)
All lags 1–256 connected in order; each row starts from its retained anchor.